

How I worked with AI on the lending portfolio analysis

Condensed English record of one Claude Code session · Sep 28–30, 2026 · 70 of my messages

About this document. The original session is long: most of my messages were in Chinese, and there are many tool and code logs. This is a condensed English version organized by project phase.

- **My messages** are translated as faithfully as I could, lightly shortened. Screenshots and uploads are described in [brackets].
- **AI replies** are summarized to the points that shaped the work.
- **Numbers** are the ones used at that point in the session; some were later refined.
- **Omitted:** a few one-word replies (for example, choosing an option) and file re-uploads.

The full, unedited transcript is available on request.

Legend: how each exchange ended

My call I set the direction, the idea or the constraint. **Refined** My question changed or corrected the AI's output. **Rejected / rewrote** I turned down the AI's approach or rewrote it myself. **Accepted** I took the AI's proposal, often after it pushed back on me. **Clarified** I asked for an explanation.

Summary of how I engaged with AI

Division of labor. I owned the scope, the storyline, the recommendations and every judgment call: thresholds, what to size, what to leave out, and how to phrase things for the audience. The AI wrote and ran the pandas code, pulled numbers, drafted the slides, and reviewed my own edits for errors. From Phase 5 on, I edited the deck myself in PowerPoint. The AI only made targeted changes to my latest file.

What I asked

- To understand the task before writing any code (#1).
- For code simple enough to explain line by line in Colab (#3–4, #27).
- "How did you get this?" for every number I would have to defend: missing FICO (#11), delinquent loss (#15), average balance (#18), 51% of loans (#34), where each slide number is reproduced (#37), seasoned vs all loans (#40), 63% (#58), 9.3 vs 9.4 (#67).
- For cuts where I had a hypothesis: repeat customers by vertical and product (#22), quarterly margins (#23), first vs second loan vertical (#39).

How I refined

- Caught data issues the AI's checks missed: the Split Pay term (#5) and a new-customer check that only covered one direction (#9).
- Challenged conclusions: an overstated missing-FICO claim was removed (#11); a funding-cost inconsistency was found (#18).
- Pushed for consistent methodology: seasoned rates on every slide (#40–41) and one sizing rule (#64–67).
- Replaced weak evidence with stronger evidence: a travel-level stress margin (#45) and the 62% vs 38% upgrade headline (#59).

What I accepted

- The root cause of four data issues: 160 corrupted duplicates (#2).
- Delinquent credit_loss as an expected loss, plus a stress case (#15).
- Pushback on deleting the users-table check; I kept it (#10).
- Keeping GMV as the term (#19–21); leaving 0% APR out of the totals (#62); showing only deltas for travel margin (#67).

What I rewrote or rejected

- Rejected the AI's risk-score model (AUC 0.68) and replaced it with a transparent what-if (#26).
- Cut 0% APR sizing from the code; it lives on the slides only (#27).
- Set my own sizing logic: the number of new merchants needed for +10% / +20% (#25).
- Reordered the deck around my storyline (#29) and added the product slide as evidence for repricing (#36).
- Wrote and edited the slide text myself; the AI then reviewed it (#43, #47–49).

Where the AI was wrong, and how it was caught

AI error	How it surfaced	Fix
Simplified loading code dropped the date conversion	My Colab run failed (#6)	Explicit conversion plus my dtype check (#7)
Claimed missing-FICO loans had "lower" charge-offs	I asked how it was inferred (#11)	Tested: not significant ($p = 0.13$); now "not riskier"
Split Pay funding cost used full principal, unlike other products	I asked why we use the average balance (#18)	Average balance for all products
"51% of loans" on a slide was computed by hand, not printed by the notebook	I asked where it was in the code (#34, #37)	Added notebook cells; all slide numbers now trace to code
Framed 145 conversions as "5% of repeat loans"	My questions on the repeat row (#56–61)	Restated as 17% of the 840 repeat Split Pay loans
Over-built Part 3 with a trained model	I judged it too complex to explain (#26)	Replaced with a what-if on the existing model

Timeline

Phase	Msgs	Main output
0 · Understand the task	1	Plan; first profile of the data; two analytical traps
1 · Data integrity	2–16	notebooks/01_data_integrity.ipynb: checks, cleaning, assumptions
2 · Portfolio & risk	16–23	notebooks/02_portfolio_risk.ipynb: unit economics, segments, repeat customers
3 · Recommendations & sizing	24–28	notebooks/03_recommendation_sizing.ipynb: three sized moves plus 0% APR
4 · Deck build	29–46	First deck, storyline, product slide, seasoned-rate methodology
5 · Review of my edited deck	47–69	Targeted fixes, repeat narrative, stress case, Appendix E and divider

The conversation, condensed

Phase 0 · Understanding the task

Sep 28

#1

ME I'm interviewing for this role [job link]. The next step is a take-home assignment with instructions and a dataset. Can you help me understand the task first? [attached: assignment PDF, loans.csv, users.csv, merchants.csv]

AI Summarized the deliverables and grading criteria, then ran a first profile of the data: duplicate loan IDs, dates outside the data window, and users-table counts that don't match the loans. Gave first signals by vertical (travel: highest yield and highest loss; fashion: most loans, thin margin). Warned about two traps: young loans make losses look too small, and declined applicants are not in the data. Proposed a 5-step plan and suggested that I make the assumption calls myself.

Accepted *Adopted the plan and the 'young loans' warning, which later became the seasoned-loan rule.*

Phase 1 · Data integrity (notebook 01)

Sep 29

#2

ME OK, I want to start step 1. I'm using Google Colab.

AI Built the first notebook and ran it end-to-end. Main finding: the 160 duplicate IDs and the 160 out-of-window dates (1923–25, 2031) are the same rows; each pair differs only in origination_date. Dropping the bad copy also fixes the users-table mismatches. Asked me to decide the seasoning cut-off and how to treat delinquent losses.

Accepted *One root cause for four symptoms became the headline of the data-quality appendix.*

#3–4

ME [pasted the AI's file-finding / upload code] Can this step be simplified? ... Thanks. I want the code to be as simple as possible, so it's easy to explain.

AI Replaced the glob/upload logic with three read_csv lines. Rewrote the whole notebook: removed the helper log() function, the issue_log table and the chart styling; one check per cell; conclusions in markdown.

Rejected / rewrote *I rejected the over-engineered first version. 'Simple enough to explain' became the standard for all three notebooks.*

#5

ME [screenshot of product × term crosstab] The PDF says Split Pay is paid in 4 biweekly installments, so the term should be about 8 weeks. But every Split Pay loan in the data has a 1-month term.

AI Confirmed the mismatch: '1' is a product code, not a real term. Corrected my estimate: the first payment is at checkout, so the plan lasts about 6 weeks, not 8. Added the assumption 'Split Pay = 1.5 months' for funding cost. Following this thread, it found 134 'current' Split Pay loans that are already fully repaid, and 21 loans flagged delinquent before they were old enough to be.

My call *I caught this by checking the product description against the data. It led to a new assumption and two more data issues.*

#6–7

ME [pasted AttributeError: .dt accessor on origination_date] ... What if we add a check here for which date columns aren't in date format, and then convert them?

AI Admitted its own simplified snippet had dropped the date conversion. Added my check: print dtypes, convert with `to_datetime(errors='coerce')`, then count values that failed to convert (0 in every column). Framed it as two layers: format (all valid) vs content (1923/2031 dates are valid but impossible).

My call *The AI caused the bug; the dtype check was my idea.*

#8

ME [screenshot of `users.describe()`] There are a lot of NaNs here.

AI Explained that the NaNs are not missing data: `describe(include='all')` mixes column types, and each type has different statistics. The 'count' row shows no missing values. Split the output into four small, readable cells.

Clarified *Accepted. The output is now easier to present.*

#9

ME [screenshot] The new-to-Affirm check may be wrong here. For someone with several loans, only the first loan should be marked new.

AI Agreed that the existing check only covered one direction. Rewrote it to compare every loan: `is_new_to_affirm` vs 'is this the user's earliest loan'. Before cleaning: 133 loans flagged new but not the first loan, and 19 first loans not flagged. After dropping duplicates: 0 and 0.

My call *My catch made the check two-directional. It also validated the new/repeat split that slide 5 relies on.*

#10

ME Can I delete this part? [the `total_loans / first_loan_date` checks in `compare_to_users`]

AI Recommended keeping it. It is the only evidence that dropping duplicates fixes the users table (150 → 0 and 53 → 0 mismatches), and it checks different fields from the new-customer flag check.

Accepted *I accepted the pushback and kept the checks.*

ME [screenshot of missing-FICO section] How did you infer a 'lower charge-off rate'? By comparing to the vertical average?

#11

AI Clarified what the claim meant, then tested it: the gap was 7 vs 11 loans and not statistically significant ($p = 0.13$). Rewrote the conclusion from 'lower' to 'not riskier' and compared the two groups' outcomes directly.

Refined *My question exposed an overstated claim, which was removed.*

ME I want to put all the assumptions in their own part (5b–5f); sections 3 and 5a count as data cleaning. ... [chose option B: AI restructures the notebook] ... Put 'issues flagged but kept' under the data-quality checks in section 2, and write the resolution in section 3. I want to say we can't be sure whether the loan age or the delinquency flag is wrong, and since the count is small, we leave them unchanged.

#12–14

AI Agreed with the split between cleaning (data is wrong; fix it) and assumptions (data is fine; choose a method). Flagged that one section mixed both categories and that the 0% APR fee finding was neither. Rebuilt the notebook as: checks → cleaning → assumptions → early observation → summary. My reasoning went into the 'Decision' text almost word for word.

My call *The notebook structure and the reasoning for keeping the 21 loans are mine.*

ME [screenshot of 4.2] Is there a way to see how the loss on delinquent loans is calculated?

#15

AI Reverse-engineered it: $\text{credit_loss} \div \text{unpaid principal}$ is 100% for charged-off loans, ~60% for 60-day delinquent and ~30% for 30-day delinquent. So for delinquent loans it is an expected loss (like a reserve), not a realized loss as the data dictionary says. These estimates are ~32% of all losses. Proposed a stress case in which every delinquent loan charges off.

My call *My question produced a key assumption. The stress case is used later on slides 3, 6 and 7.*

ME The early observation on the 0% APR merchant fee can wait until later. Let's start Part 2.

#16

AI Moved the 0% APR observation to the Part 2 product analysis.

My call *Sequencing decision.*

Phase 2 · Portfolio health & risk (notebook 02)

Sep 29

AI Built notebook 02 around one helper, `summarize(df, by)`. Findings: travel FICO<640 has a ~40% bad rate; fashion earns ~10% of profit on half the loans; healthcare and home furnishing look best. Changed funding cost to the average balance and asked me to confirm. Proposed a direction: invest / tighten / reprice.

#16 (cont.)

Accepted *Accepted the direction as a starting hypothesis.*

ME Can we just call the cleaned table `loans_clean`, so it doesn't cause confusion?

#17

AI Renamed it in both notebooks. Kept the `compare_to_users(df)` parameter neutral, because it runs on both the raw and the clean table.

My call *Naming for readability.*

ME I also want to ask: why do we use the average balance?

#18

AI Installments amortize, so on average only about half the principal is outstanding. Using full principal would roughly double funding cost (2.1% vs 1.1% of GMV). Showed that the exact factor is $(n+1)/2n$ and that the vertical ranking is unchanged under all three methods. My question also surfaced an inconsistency: Split Pay was still on full principal. Fixed it for all products.

Refined *A 'why' question found an inconsistency in the AI's own method. I can now defend the assumption.*

ME Could we avoid 'GMV' and just say loan amount / total loan amount? ... [interrupted] Never mind, keep GMV. I hadn't noticed the data dictionary defines it explicitly; I was just worried it would cause confusion.

#19–21

AI Stopped, reverted the rename and verified that the rebuilt notebooks match the committed ones.

My call *I reversed my own request after re-reading the dictionary.*

ME For repeat customers, I'd like to analyze by vertical and by product in the health section: which one generates more repeat customers? In the risk section I also want repeat vs non-repeat loss by vertical and by product.

#22

AI Used a fair cohort: first loans made by Jun 2025, repeat within 180 days. Repeat behavior is driven by vertical, not product: fashion, fitness and electronics are at 94–96%; big-ticket verticals at 17–23%. 63% of Split Pay starters who return take an interest-bearing loan next. Repeat loans lose far less, and 82% of losses come from first loans; this holds on seasoned loans. Noted that this changes the fashion story: fashion is an entry channel.

My call *My hypothesis. It became slide 5 and the 'repeat customer' investment.*

ME [screenshot] In portfolio health, I'd also like margin by vertical and by product for each quarter.

#23

AI Warned that recent quarters are immature. Split the trend into real vs artifact: the fall in interest-bearing margin is immaturity (interest not yet collected), while the rise in Split Pay loss is real (the product fully matures in ~6 weeks). Corrected two of its own overstated sentences after checking the table.

My call *My request. It became Appendix C.*

ME Can we move to Part 3? Does Part 3 still need code, or can we go straight to slides?

#24

AI Said a small amount of code is needed to quantify impact, as the brief asks. Drafted five recommendations and asked me to make three judgment calls: how to treat travel FICO<640, the growth scenarios, and the base year.

Accepted *Agreed to keep a short sizing notebook.*

ME Rec 1: let's look at how many consumers and how much profit one healthcare or home furnishing merchant brings per year. If we want +10% GMV, how many new merchants would we need? And for +20%? Rec 2: I agree with your quantification. Rec 3: at a high level, lower the approval rate or the credit limit for FICO<640; the model can decide this from the risk score it generates from the customer profile.

#25

AI Built my merchant-based sizing: ~57 / ~122 consumers and ~\$10K profit per merchant, so +10% / +20% needs 0.3–0.8 new merchants per vertical. For Rec 3 it went further than I asked: it trained a risk-score model (logistic regression, AUC 0.68 vs 0.60 for FICO alone) and sized an 'approve half the limit' policy.

My call *The sizing logic for Rec 1 is mine.*

ME 'Tighten approvals for the riskiest FICO<640 applicants using a risk score' is too complex. I think this belongs on the slides; I don't need to build a model. What I meant is that this can be done by adjusting the parameters and targets of the existing model.

#26

AI Removed the model and the sklearn dependency. Replaced it with a transparent what-if: the existing risk model is tuned so that segment volume falls 25% / 50%, and profit and loss scale with it.

Rejected / rewrote *I rejected the AI's model. Explainability matters more than modeling for this audience.*

ME This change works. We only do assumptions and sizing; explainability is what I care about most. Reprice 0% APR can also go straight on the slides, since the earlier numbers are already clear. And please simplify this code where you can.

#27

AI Dropped 0% APR from the notebook. Rewrote each recommendation as one small function returning {gmv, profit, loss}, and replaced positional iloc lookups with named totals. Result: GMV +3.9% / +7.7%, profit +7.4% / +14.8%, loss rate 2.32% → 2.21% / 2.10%.

Rejected / rewrote *I cut scope and rewrote for readability; the numbers now trace line by line.*

ME Yes, you can use the brand colors, but don't write the company name.

#28

AI Built a 7-slide deck plus appendix in an indigo/navy palette with no name or logo. Titles state conclusions; speaker notes are drafted. Suggested I rewrite the notes in my own words.

My call *Presentation constraint.*

Phase 4 · Building the deck

Sep 29

ME I want to change the structure to 3-4-5-2-6-7: first the state of the portfolio, then the drivers, then the investment plan and trade-offs.

#29

AI Reordered the slides and relabeled 'Recommendation' as 'Investment plan'. Suggested putting a one-line answer on the cover so executives still hear the conclusion first.

Rejected / rewrote *I rewrote the storyline: situation → drivers → plan → trade-offs.*

ME [screenshot of slide] I'd like this chart to have a secondary y-axis in percent with three lines: revenue, credit loss and funding cost as % of GMV.

#30

AI Built it and noted that dual-axis charts can mislead. Flagged that the revenue line drops in the last two quarters because those loans are immature, and added a footnote so reviewers don't ask 'why is revenue falling?'.

My call *My chart design; the AI added the caveat.*

ME Can't you generate Google Slides? ... Can I connect a Google account or Drive? ... Never mind, I've downloaded PowerPoint.

#31–33

AI Explained that the session had no Google connection and listed conversion options.

My call *Tooling choice: I edit the deck in PowerPoint myself from here on.*

ME '51% of loans but 10% of profit': what does 51% of loans mean, and where can I find it in the code? ... No need to change the code; just add a 'share of loans' column before share of GMV on the slide.

#34–35

AI Explained that it is the share of loan count (3,621 of 7,134), and admitted it had been hand-computed, not printed by the notebook. Added the column to the slide.

Refined *A traceability question exposed a number that wasn't reproducible.*

ME I think we also need a by-product chart like the one on slide 3; otherwise we can't reach the 'reprice' conclusion.

#36

AI Added a product slide: interest-bearing loans earn 82% of profit, while 0% APR gives up interest without a higher merchant fee (5.1% vs 5.5% at the same merchants).

My call *I required evidence before the recommendation.*

ME Where on GitHub are the slide 3 and 4 numbers reproduced? ... OK, add them.

#37–38

AI Mapped every number to a notebook cell and found three that were not reproduced (share of loans, and the fee/interest split). Added two small cells; outputs match the slides exactly.

Refined *Every slide number is now traceable to the code appendix.*

ME [screenshot] I'd like to see which vertical these repeat customers started in and which vertical their second loan was in. Also, slides 6 and 7 can be merged.

#39

AI 86% of second loans stay in the same vertical, so cross-selling from fashion into big-ticket verticals is small. Added a card, Appendix D (a 7×7 heatmap) and a notebook cell. Merged the plan and the sizing into one table.

My call *My question showed repeat growth must be driven within each vertical.*

ME Do slides 3 and 4 only use seasoned loans? ... [chose B]

#40–41

AI No: they used all loans. Offered A (keep all loans and add a robustness note) or B (rate metrics on seasoned loans, share metrics on all loans) and recommended B. Implemented B across the slides, the notebook and the methodology appendix.

Refined *My question made the methodology consistent across the deck.*

ME '...yet at the same promo merchants its fee averages 5.1% vs 5.5% for interest-bearing': what does this mean? ... Never mind, I rewrote it myself. On slide 3, next to '~43% delinquent or charged off', I want to add the bad/loss rate for low-credit-score customers in other verticals as a comparison, and briefly say why I chose 640 as the threshold.

#42–43

AI Explained the merchant-fee subsidy logic and verified that the gap holds at every one of the 15 promo merchants. For slide 3: FICO<640 in other verticals has a ~7% bad rate vs 43% in travel. Travel's bad rate steps down at 640 (36–47% below vs ~21% at 640–659). Flagged a typo in my draft (7.7% should be 7.1%).

Rejected / rewrote *I wrote the slide text myself; the AI supplied the supporting numbers and a typo check.*

ME [screenshot] I don't think 'Margin 7.7% falls to ~1% if delinquent loans charge off' is very convincing. How about saying how the whole travel category's margin would fall?

#45

AI Reframed at category level: under the stress case, travel margin falls from 9.4% to 5.2%. FICO<640 is 35% of travel GMV but 64% of travel losses.

Rejected / rewrote My reframing gave a stronger argument: tighten one segment rather than exit travel.

ME Along the same lines, I want to update the 'why' for travel FICO in the investment plan, and make the reasoning in 'how' clearer. For growing healthcare & home furnishing, I want to say how many customers one merchant contributes and what one customer's loan earns. For repeat customers, it should mainly be marketing initiatives that convert Split Pay to interest-bearing: what does each conversion bring?

#46

AI Computed per-unit economics: ~\$176 / ~\$78 profit per loan, ~\$10K per merchant, and +\$49 profit per Split Pay → interest-bearing conversion. Rewrote the 'how' cells in these units.

My call I set the explanation format: per merchant, per loan, per conversion.

Phase 5 · Reviewing my own edited deck

Sep 30

ME [uploaded my hand-edited .pptx] Just sync the change into this file. ... [uploaded again] Done. For the slide 3 fashion card, margin can just use all loans. Can you check whether anything else is wrong? ... Is it right now?

#47–49

AI Changed only the requested slide and verified the rest of my file was untouched. Reviewed my edits: two swapped rows on slide 3, a mistyped figure ('2.38' should be 2.8%), outdated numbers in the speaker notes, and mismatched notebook references. Then re-checked slide 3 cell by cell.

Accepted From here on I own the deck; the AI works as a reviewer and makes targeted edits on my latest file.

ME On this page [Appendix A] I want to add the new-customer flag as its own row. Did we adjust for this issue? ... What do '133 + 19' and '6.8%' mean?

#50, 53

AI Explained that the issue was resolved by dropping the duplicates (133 + 19 wrong flags → 0), added the row, and explained both numbers with their notebook locations.

My call Documented a fix that had been happening silently.

#54

ME The cover says 'Grow low-loss verticals and repeat customers; tighten the riskiest travel credit.' Do we need to say low-loss and high-margin?

AI Yes: the choice rests on both, and different stakeholders care about each (risk vs merchant partnerships). Suggested 'high-margin, low-loss'.

Accepted Wording accepted.

#55

ME [screenshot of slide 6] I'm still a bit confused about GMV / profit / loss on this page. The conservative and ambitious calculations aren't shown.

AI Walked through every row: 2025 volume × assumption × seasoned unit economics. The ambitious case doubles each assumption. Suggested a new appendix page with the math.

My call My confusion was a signal that reviewers would be confused too. It became Appendix E.

#56–61

ME What does this mean: 'Marketing offers that convert repeat Split Pay customers to interest-bearing...'? ... I want to write it as: convert Split Pay (entry) into repeat customers, and also invest in repeat customers. ... Is 63% the share of all Split Pay [customers] who repeat and then take interest-bearing? ... 63% is fine, but on slide 5 I want to stress that repeat customers choose interest-bearing the most. ... Then please improve the 'invest in repeat' row. ... So 'upgrade' targets customers who already repeat but still repeat on Split Pay?

AI Clarified the denominators: 63% is among Split Pay starters who return; 54% of all eligible starters. Found the cleaner headline for my point: 62% of second loans are interest-bearing vs 38% of first loans. Rewrote the row as two actions, Upgrade (840 repeat Split Pay loans, +\$49 each) and Bring back (big-ticket repeat rate, ~\$126 per loan). Also corrected its own framing: 145 conversions = 17% of repeat Split Pay loans, not 5%.

Refined My narrative, Split Pay as the entry point that upgrades on return, sharpened through five rounds of questions.

#62

ME Why doesn't Total (moves 1–3) include move 4?

AI Repricing depends on how merchants respond, the \$1.9K is upside only, and the effect is small (+0.9% profit). Showed what the totals would be if it were included.

Accepted Kept it out of the totals and prepared the interview answer.

#63

ME For tighten, profit looks like it falls, but the margin actually improves. How can we show that?

AI Checked first: portfolio margin barely moves (7.43% → 7.44%), so 'margin improves' doesn't hold at portfolio level. But travel margin rises (+0.2 / +0.5 pts), travel loss falls, and under the stress case the profit given up is only \$0.3K / \$0.7K.

Refined My intuition was partly right. The AI corrected the framing, and I used the stress-case line instead.

#64–67

ME For the impact I'd keep the calculation the same as before: volume can use all loans, but rates should use seasoned results ... [interrupted] Oh, got it, full-year 2025 makes sense. ... Is travel's seasoned margin 9.3 or 9.4?

AI Explained that 9.4% is the observed seasoned margin and 9.3% is the same rates reweighted to the 2025 mix. Recommended showing only the changes (+0.2 / +0.5 pts) so the two slides don't conflict.

Accepted *One consistent sizing rule, and no conflicting numbers across slides.*

#68–69

ME [screenshots of slide 7] Can you check what needs changing here? For example, 'Test second-loan offers for big-ticket first-time customers' needs rewording. We also need to start negotiating the 0% APR merchant fee, but in a way that keeps merchants from dropping out. I'd also like to add a page showing the impact calculation, and a separate 'Appendix' divider page.

AI Aligned the next steps with the four moves. Proposed a merchant-friendly repricing approach: start with the merchants furthest below parity, tier fees by promo length, back the ask with promo-lift data, and offer co-marketing in exchange. Added the stress-case trade-off, the Appendix divider and Appendix E (impact sizing).

My call *The requirement not to lose merchants while repricing is mine; the AI drafted the approach.*